



Endless Pac-Man

Pac-Man turned 35 recently and Nintendo's released a new title to celebrate. You can now play endlessly <http://dgit.in/endlesspacman>

Outlook for iOS updated

Outlook users can now open their MS Office attachments easily and edit them. <http://dgit.in/newiosoutlk>



Advantech demonstrating an intelligent smart ticketing system on the road

human intervention. On its roof, sits one of the largest privately owned solar-panel installations in Taiwan, powering the building's electricity needs. It has touch-enabled smart displays (running on Windows 7 no less) throughout the building, allowing residents and visitors to interact with it and get information. Conference rooms with internet-connected motion sensors let you know if a room is occupied by automatically turning red or green on its outside.

Whether it be automatic fare collection on the road, smart parking spaces, intelligent buildings or the next wave of smart factories that will help us reduce the adverse impact of technology on our ecology, Advantech's innovations are ensuring IoT isn't just a pipe dream anymore. It's more than real.

Chip-level innovations: ARM and MediaTek

As you probably know, ARM chips are based on "reduced instruction set computing" or "RISC". This provides ARM devices with significant gains in the area of power-efficient computing – of paramount importance to sensor end-points, especially ones that need to last quite long (sometimes several years) with really low-powered batteries. And here's how ARM will ensure that IoT becomes an end-to-end reality.

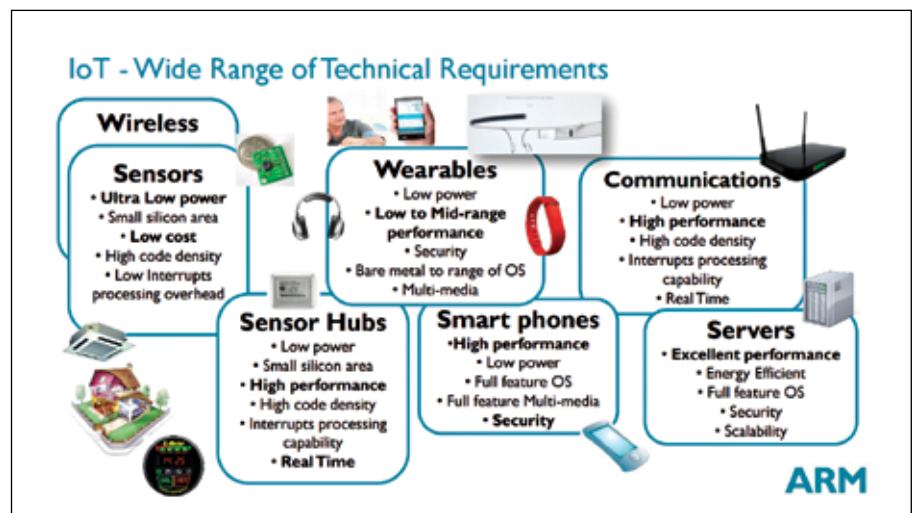
In ARM's family of processors, the Cortex-M is what it thinks will drive the IoT trend across the globe, since by its very definition the Cortex-M family of processors are used primarily for computing tasks involving smallest footprint and lowest power consumption. Things like heart pacemakers, smart street lights, smart washing machines, and wearable devices, among other things, are powered largely by ARM's Cortex-M processors. Whether it's deeply embedded wearable devices like the Nike Smart Shoe or Fitbit Flex or smartwatches like Moto 360, right from measuring 90 nanometres to lasting for up to 15 years, ARM Cortex-M

processors are at the heart of the wearable devices phenomenon.

But that's not enough, is it? ARM has two initiatives specifically targeting the IoT wave in the form of 'ARM Cordio Radio IP' and 'ARM mbed OS'.

Connectivity is the name of the game, and ARM is trying to push the envelope on this by trying to reinvent radio communication standard at the silicon level. The Cordio Radio IP series includes integrated, ultra-low power CMOS radio IP cores along with link layer firmware, stack and profiles supporting 'Bluetooth Smart' – a new standard that sets the stage for a variety of new applications needing ultra-low power wireless connectivity at low cost. This technology is particularly suitable for devices that need to transfer small amounts of data within a relatively short range (2.4GHz frequency, throughput < 305 kbps, range < 50 m). Like existing Bluetooth technology, Bluetooth low energy is universal, interoperable and license-free. Why is this necessary? Because traditional Bluetooth and wireless data standards are too taxing in terms of data packet and power consumption, both of which are managed more efficiently in Cordio Radio IP.

ARM's mbed platform enables device manufacturers and developers to optimally configure chip-level processes. With a chip-level, local, configurable mbed OS (to collect little data) and Internet-connected mbed device server (to analyse big data), IoT application or services can be



ARM's chips are powering end-to-end hardware solutions in the IoT era